

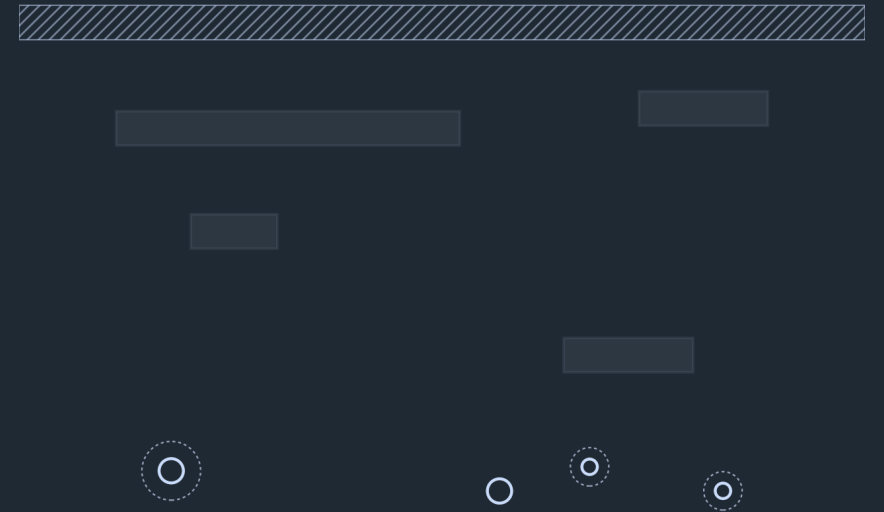
# CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

Lavinia Pedrollo<sup>1</sup> • David Gvazabia<sup>2</sup> • Torben Graeber<sup>3</sup> • Martin Fischer<sup>1</sup>

<sup>1</sup> Stanford University, CIFE   <sup>2</sup> Lafayette College   <sup>3</sup> Hilti AG

43rd International CIB W78 Conference • IT in Construction • New Delhi, 6–8 October 2026



a generated C8 context: eight services on three rows

# Every pipe, duct and tray hangs from a support designed by hand

$\approx 10,000$

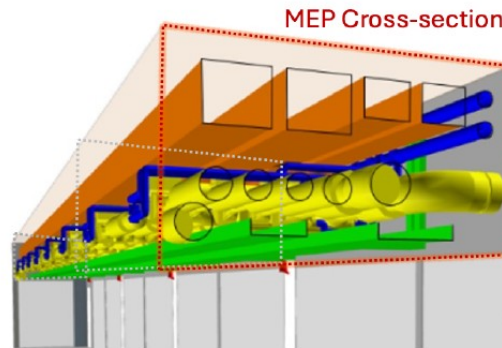
support assemblies in one 200,000 sq ft hospital

**20 min – 2 h**

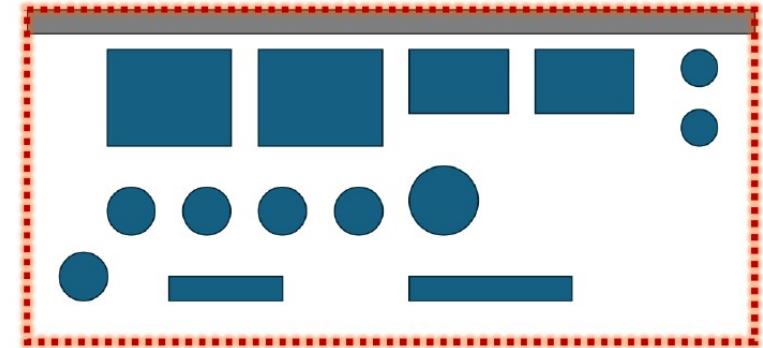
to design each one by hand, from the coordinated model

$\approx \frac{1}{4}$

of all MEP design effort; on the order of \$600K of engineering per project



(a) Full MEP route



(b) One CrossMEP context

Support design starts after coordination. At each hanger location the designer works from the section across the run: what passes, how big, how heavy, how far apart, what it hangs from.

*Nobody designs a support from the whole model; it is designed from the section at that hanger.*

**That section is the unit of work — and the unit of CrossMEP.**

Approximate practitioner estimates.

# There was no public dataset of support-design problems

## Why it has to be synthetic



### Scarcity

Project models are proprietary; the few open ones are not organised around supports and often export no element sizes.



### Coverage

One project is one narrow slice: one building type, one trade mix. A method tuned to it measures fit to that project.



### Control

A seeded generator gives unlimited problems with a known distribution, difficulty by construction and held-out seeds.

## CrossMEP at a glance

**7,000**

contexts in four splits  
on disjoint seeds

**31,484**

elements: pipes, cable  
trays, ducts and  
conduits; six trades

**C1–C8**

eight difficulty tiers by  
exact element count

**0 labels**

the brief, not the  
answer: no support-  
assembly information

**Traced**

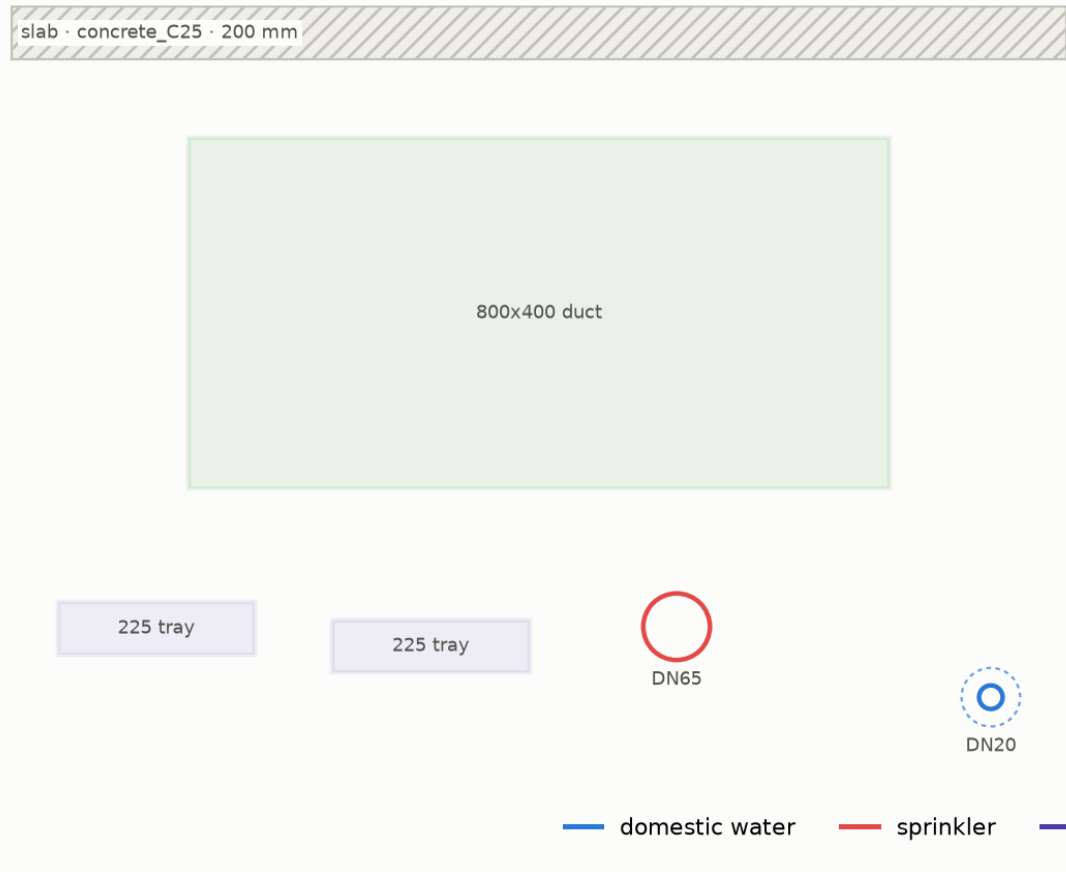
every constant with its  
source, or declared a  
design choice

**Open**

data CC BY 4.0, code  
MIT; a seeded,  
deterministic generator

# A context is the section at one hanger: the brief, not the answer

## One context (mep\_228, tier C5): the section at a support location



## What the support designer is given

| element      | trade          | ins | kN/m  | span  | kN   |
|--------------|----------------|-----|-------|-------|------|
| 800x400 duct | ventilation    | 0   | 0.240 | 2.4 m | 0.58 |
| 225 tray     | electrical     | 0   | 0.405 | 2.0 m | 0.81 |
| 225 tray     | electrical     | 0   | 0.405 | 2.0 m | 0.81 |
| DN65         | sprinkler      | 0   | 0.100 | 3.0 m | 0.30 |
| DN20         | domestic water | 20  | 0.019 | 2.1 m | 0.04 |

surface: ceiling, concrete\_C25, 200 mm

total load at this support: 2.54 kN (ins = insulation per side, mm)

bundle width 1099 mm, 2 rows, closest pair 88 mm clear

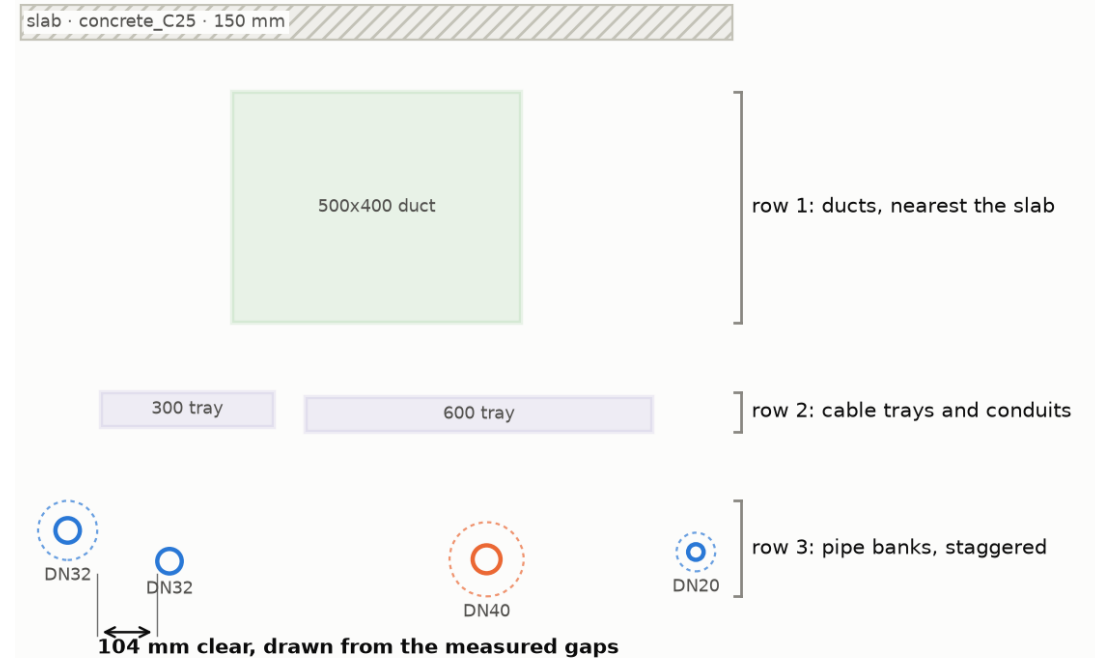
**Not given - that is the answer: channel, rods, clamps, anchors.**

Per element: kind, service and trade · bare size · insulation per side · load per metre × the span it was sized at = load at the support · position along and out from the surface. Per context: slab or wall, substrate, thickness. Absent by design: channel, rods, clamps, anchors — and any “correct” answer, since a feasible support depends on the catalog you build from.

# How a context is generated: rules, not free randomness

- 1 Tier.** Cn fixes the exact element count,  $n = 1 \dots 8$ .
- 2 Surface.** Ceiling or wall (78 / 22 %); concrete slab or wall, 150–300 mm thick.
- 3 Services.** Grouped as trades run: hot + cold pairs, flow + return, chilled banks, sprinkler mains, 1–2 trays, ducts, conduit groups of 2–6.
- 4 Rows.** Bulky first: ducts nearest the surface, then trays and conduits, then pipes.
- 5 Spacing.** Gaps drawn from gaps measured on a built project (at least 25 mm); stagger within rows; electrical above wet on walls.
- 6 Check.** Every pair at least 25 mm clear; rows centred; sizes, insulation, loads and positions recorded.

mep\_150, tier C7: 7 elements, 3.92 kN



One generated benchmark context, three rows by priority; the gap marked is one draw from the measured distribution. Every context is seeded: the same seed gives the same file, byte for byte.

# Every constant is sourced or declared a choice; every load is computed

| Element             | Sizes                                                          | Load basis                                                            |
|---------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| Pipes DN15–100      | EN 10220 / EN 10255 medium series                              | steel + water × ASME B31.1 water-service span (published points only) |
| Insulation          | GEG Anlage 8 (heated); 30/50 mm condensation control (chilled) | —                                                                     |
| Cable trays 150–600 | IEC 61537 systems                                              | full tray: 50 kg/m at 300 mm datum, × 2.0 m                           |
| Ducts               | EN 1505 rectangular; EN 1506 round                             | manufacturer duct-weight table incl. flanges, × 2.4 m                 |
| Conduits Ø20–50     | IEC 61386-1, parallel groups of 2–6                            | steel tube + 40 %-of-bore cable fill, × 2.0 m                         |

## Layout conventions

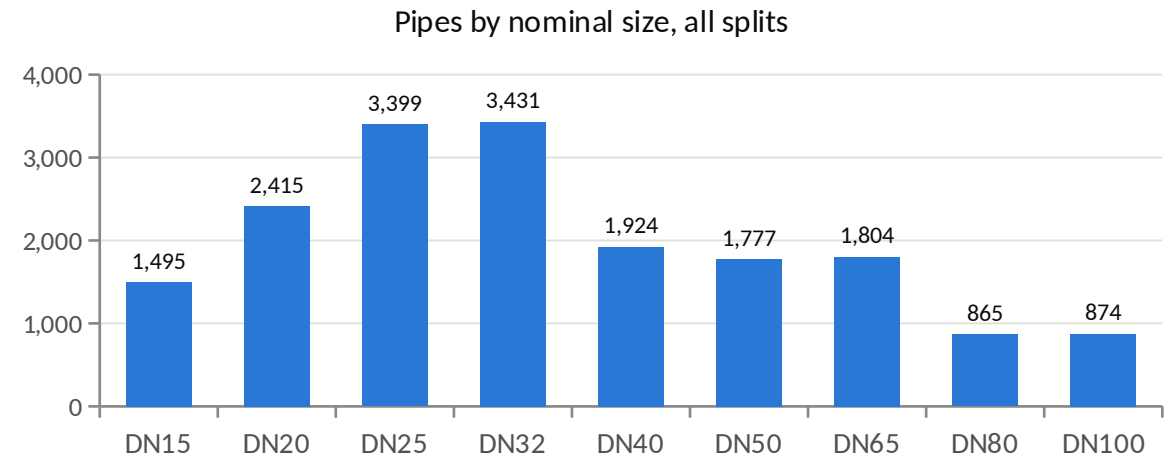
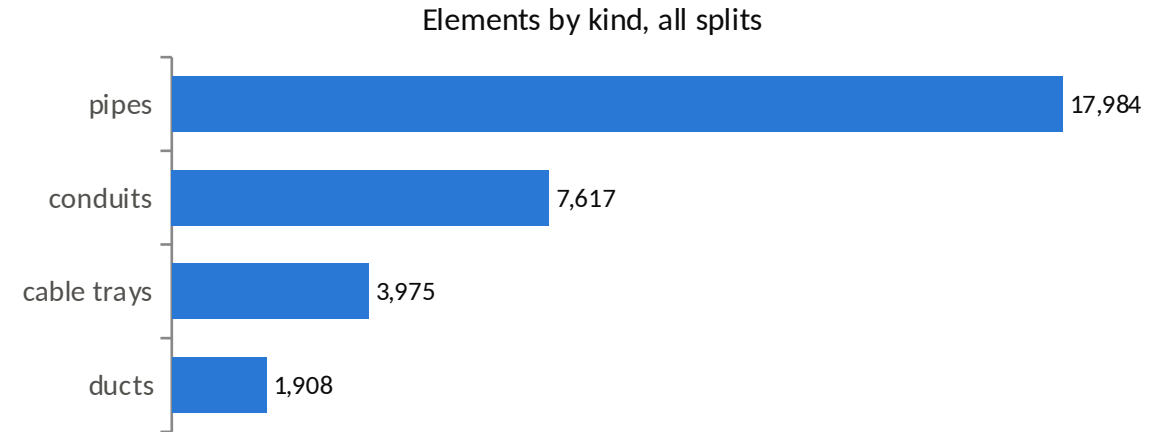
- Bulky services nearest the slab: ducts, then containment, then pipes
- Trades run in banks: hot + cold pairs, flow + return, conduit groups
- Electrical kept above wet services on walls (drip)
- Clear gaps drawn from a distribution measured on a built project; 25 mm minimum
- Within-row stagger calibrated to measured elevation spread

Every constant carries a source and a status (verified, practice-cited, or a declared design choice) in the repository. Loads are derived in code from those sources, so each number can be traced and checked.

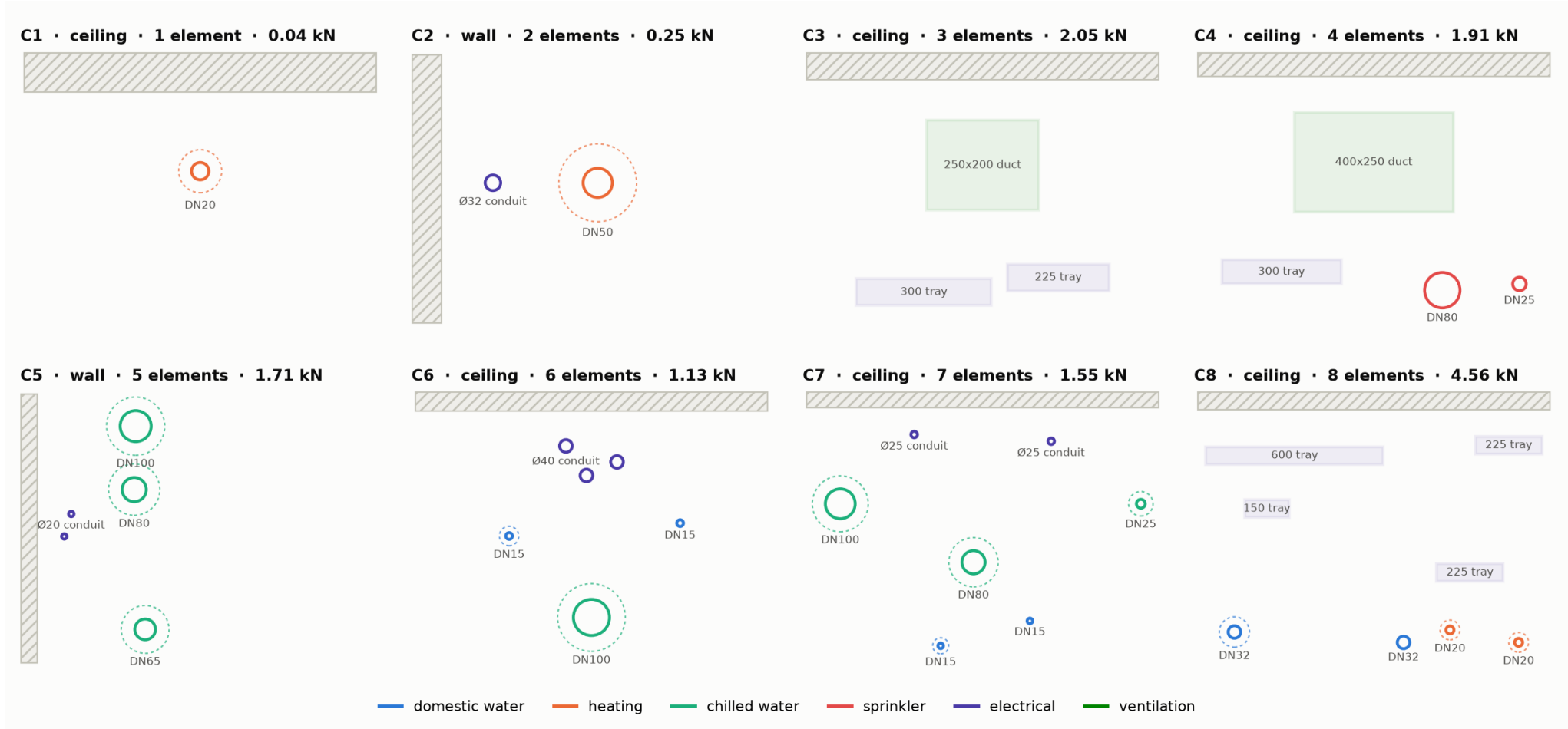
# The release: 7,000 contexts in four splits on disjoint seeds

| split        | contexts     | elements      | seed |
|--------------|--------------|---------------|------|
| train        | 5,000        | 22,500        | 1000 |
| validation   | 500          | 2,242         | 2000 |
| test         | 500          | 2,242         | 3000 |
| benchmark    | 1,000        | 4,500         | 42   |
| <b>total</b> | <b>7,000</b> | <b>31,484</b> |      |

- Tiers assigned round-robin: 625 per tier in train, 125 in the benchmark
- Four in five contexts hang from a ceiling, one in five from a wall
- Six trades: electrical 11,592, domestic water 5,889, heating 5,766, chilled water 4,129, sprinkler 2,200, ventilation 1,908 elements
- Plain JSON, with a JSON Schema, Croissant metadata and a datasheet

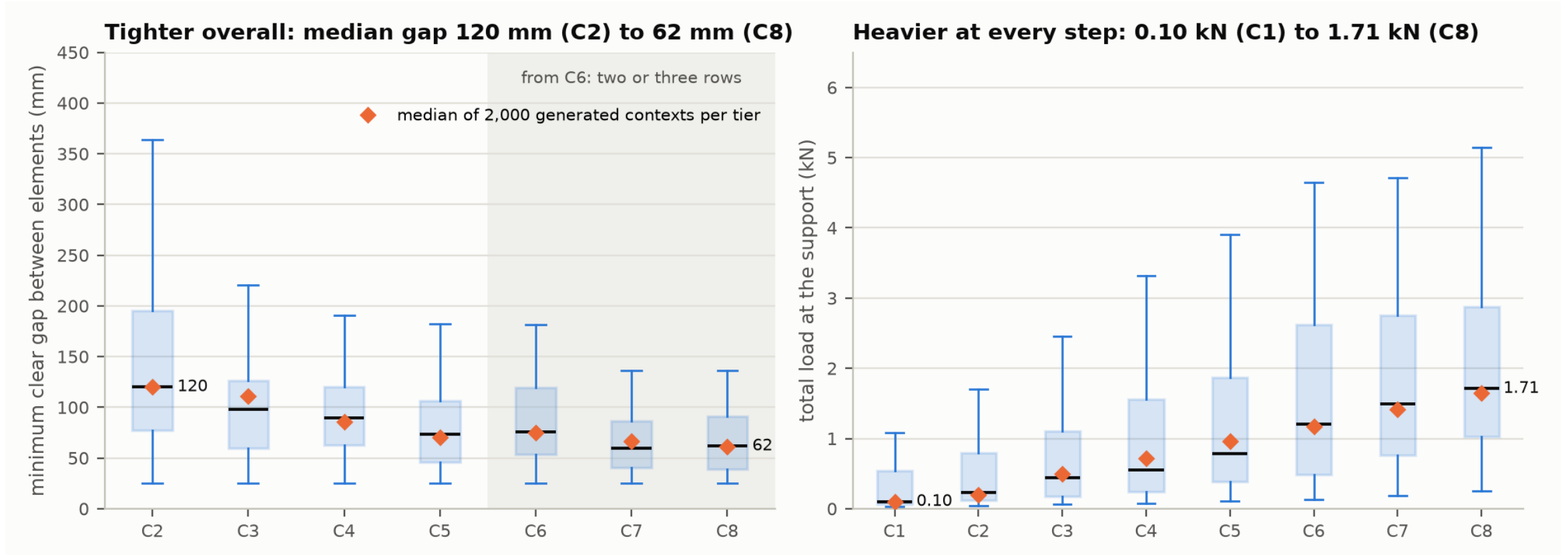


# Eight difficulty tiers: tier Cn holds exactly n elements



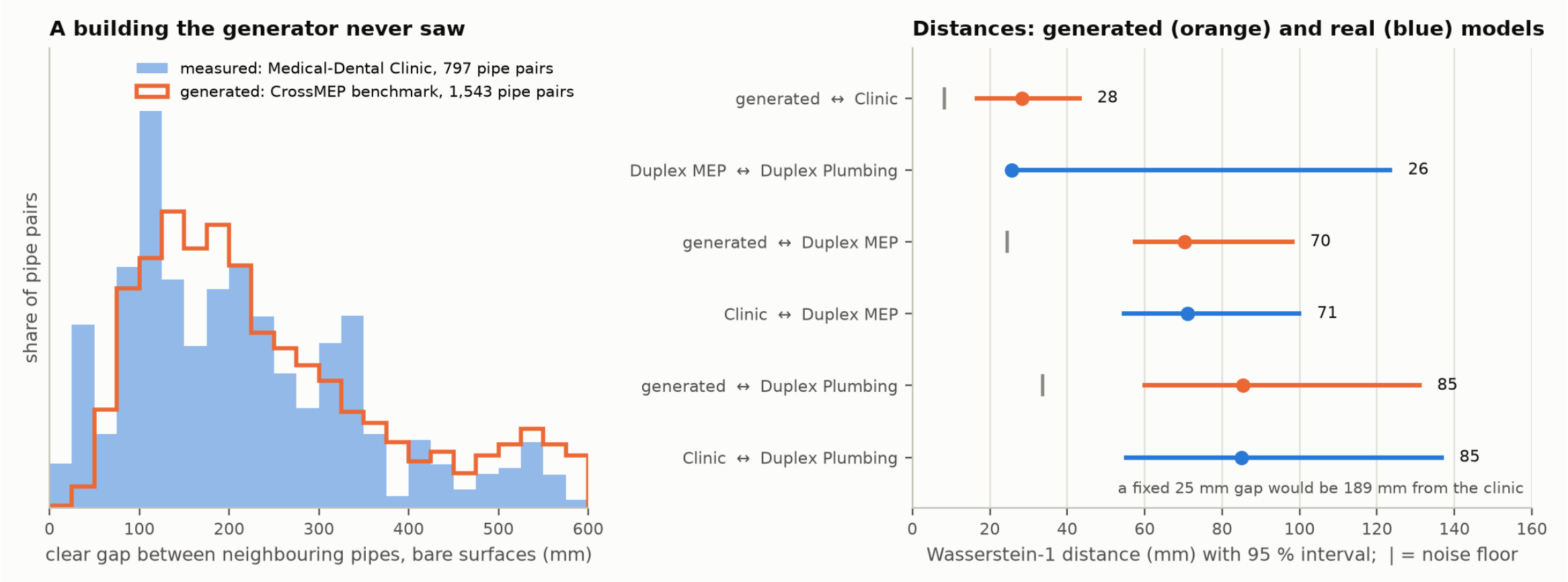
One benchmark context per tier; dashed rings are insulation. Within a tier, kinds, trades, services, surfaces and stacking all vary, so element count is the one controlled difficulty axis. C1 is the most common support in any building; C8 is a congested rack.

# Experiment 1, a design check: difficulty grows with the tier



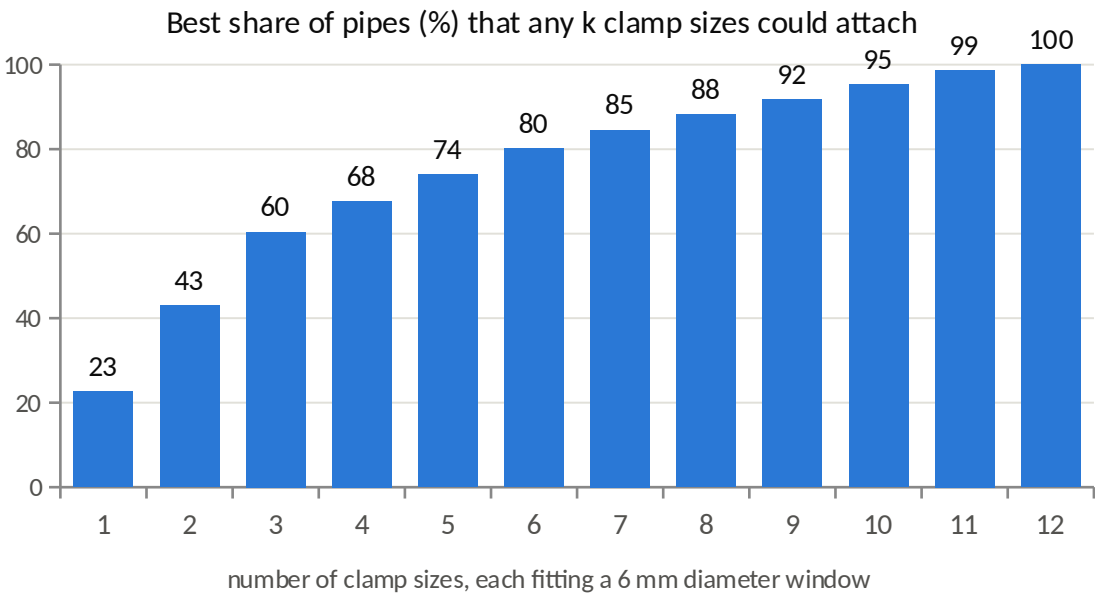
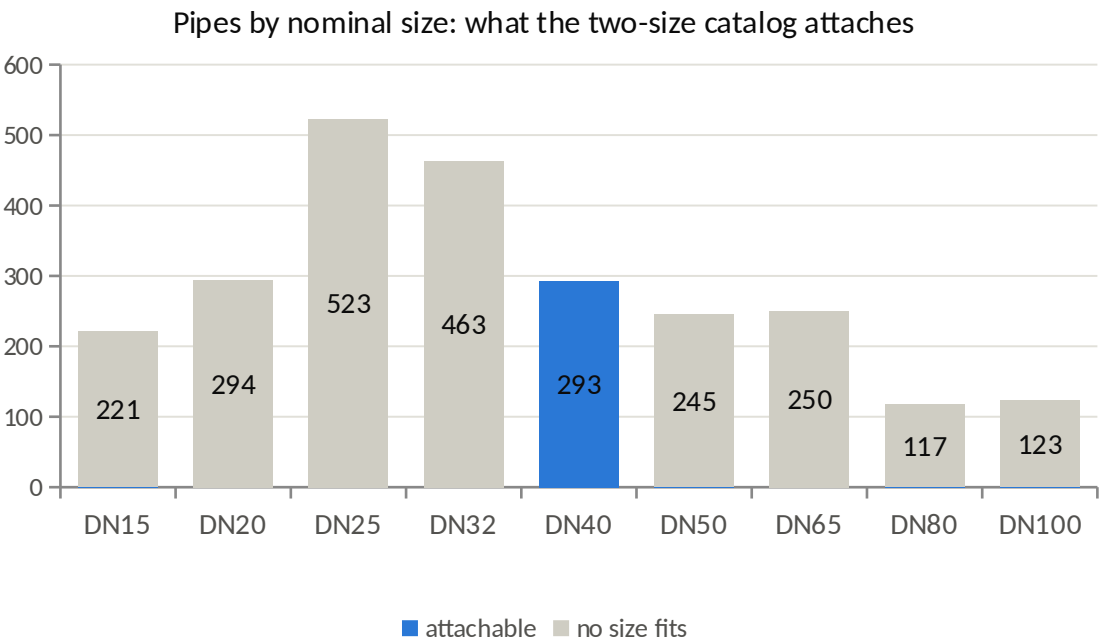
A check that the dataset behaves as designed: the tier fixes the element count; load and congestion follow from the rules. Boxes: benchmark, 125 contexts per tier (IQR, whiskers 1.5 IQR); diamonds: medians of 2,000 generated contexts per tier. Load rises at every tier (rank correlation +0.56); the closest gap shrinks overall (-0.36) but widens at C6 (70 → 75 mm), where elements start to stack in two or three rows.

# Experiment 2, the real test: spacing in a building it never saw



The gap distribution was fitted on the duplex; the clinic was held out. Clear gaps between side-by-side pipes, on sections every 250 mm. Clinic (797 pipe pairs): 28 mm from the generated gaps (95 % CI 16–44), more than the 8 mm a perfect generator would show, and about as close as the duplex’s own two models are to each other (26 mm). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0.

# Experiment 3, a use: what must a clamp catalog cover?



11.6 %

of benchmark pipes attachable with the paper's two sizes (95 % CI 10.1–13.0); every one is DN40

43.0 %

with two sizes placed where the pipes are; 80 % with six, and every pipe with 12

0 of 1,971

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (0.88 vs 2.5 kN)

# See it: set the parameters, get the support designer's brief

SERVICES AT THE HANGER

By tier Exact mix

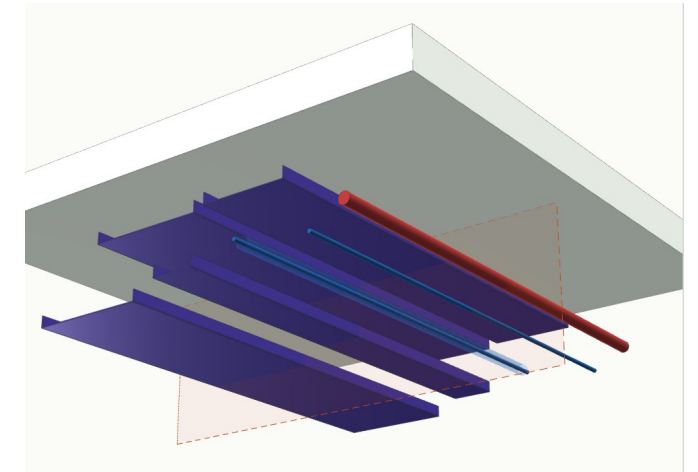
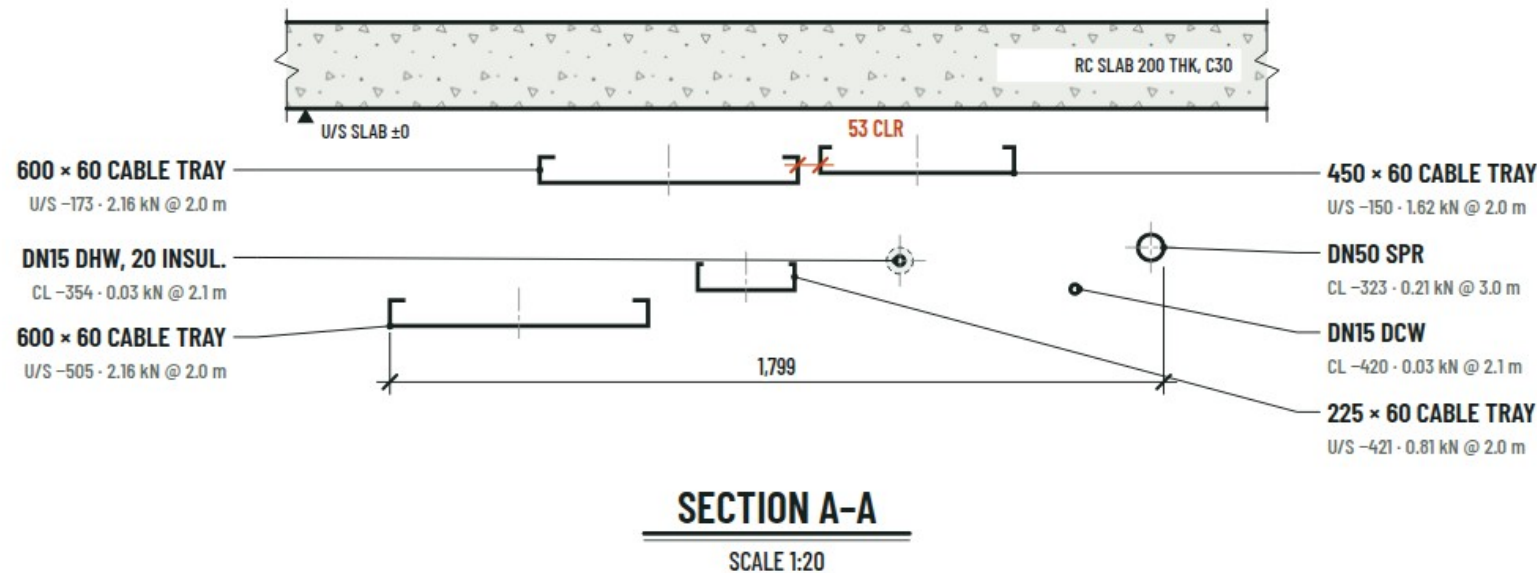
TIER = NUMBER OF SERVICES

C1 C2 C3 C4 C5 C6 C7 C8

SEED

< 0 >

Generate another



Scan to open the studio

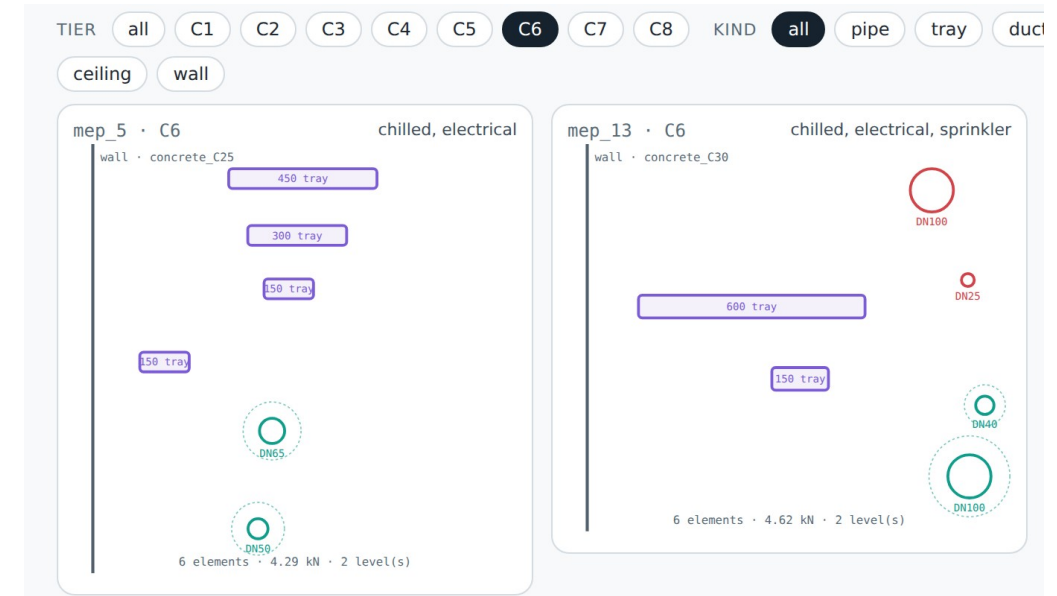
In: the parameters (tier C7, seed 0). Out: section A-A at the support, as the support designer receives it (scale 1:20), and the same run in 3D. Every section is a stored output of the released generator, with the Python call that reproduces it.

# Using CrossMEP: train, evaluate, report

```
import crossmep.tasks as cm
data = cm.load("benchmark") # 1,000 contexts
cm.filter_contexts(data, pipes=2, trays=1)

from crossmep.evaluate import score, compare
score(my_results, data) # per tier, 95 % CI
compare(mine, baseline, data) # paired test
```

Loading, metrics and scoring need only the Python standard library; the generator needs NumPy. Command line: `python -m crossmep generate | validate | evaluate`.



Interactive gallery: filter the benchmark by tier, kind, trade and surface.

## Train

on the generator: unlimited seeds, a curriculum from C1 up to C8

## Evaluate

on the benchmark: 125 held-out contexts per tier, the same for every method

## Report

per tier with 95 % intervals, paired tests between methods

# Scope, and what comes next

## Scope

- One cross-section at one support: routing, branches and elevation changes are outside it; so is support spacing — the span is recorded per element, so a method can vary it
- Realism checked on sections of two open buildings, a residential duplex and a medical clinic; congested racks rest on coordination practice and standards
- Trade mix and tier composition are design choices, not survey results
- Synthetic: not for the structural design of real installations

## Next

- A public checker for support designs, with best-known costs from an exact solver on the small tiers
- The catalog as an input: generated catalogs, and tests on catalogs a method has never seen
- A real test set: sections from commercial projects with supports designed by engineers

**If you coordinate services or design supports**, open the gallery, filter for the scenes you know, and tell us what looks wrong. Every rule is one documented constant away from being changed.



# CrossMEP: the brief, not the answer

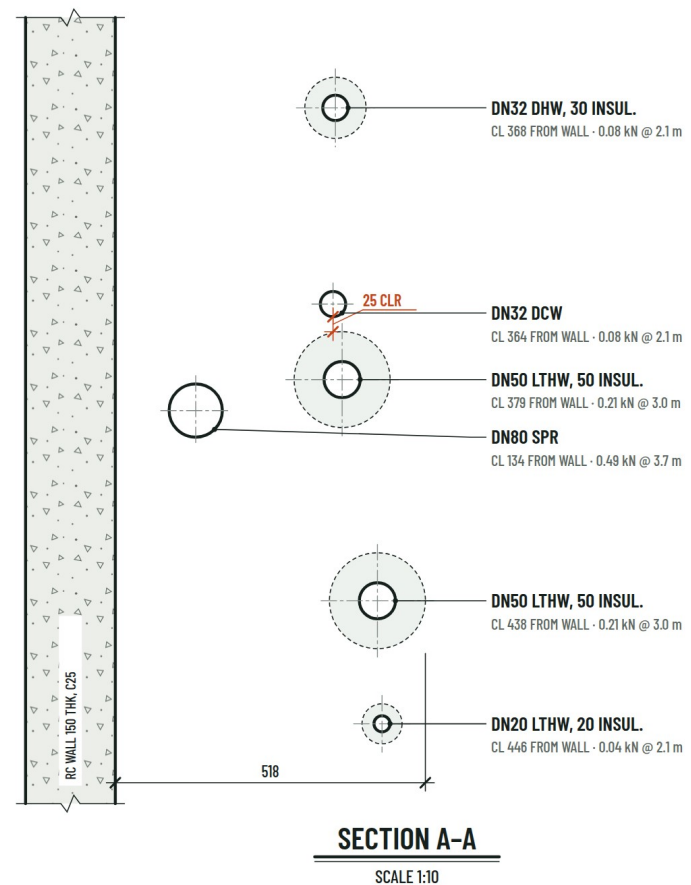
7,000 support-design problems · every constant sourced or declared  
pipe gaps checked on two open buildings · open data and code

Data CC BY 4.0, code MIT. Not for the structural design of real installations.

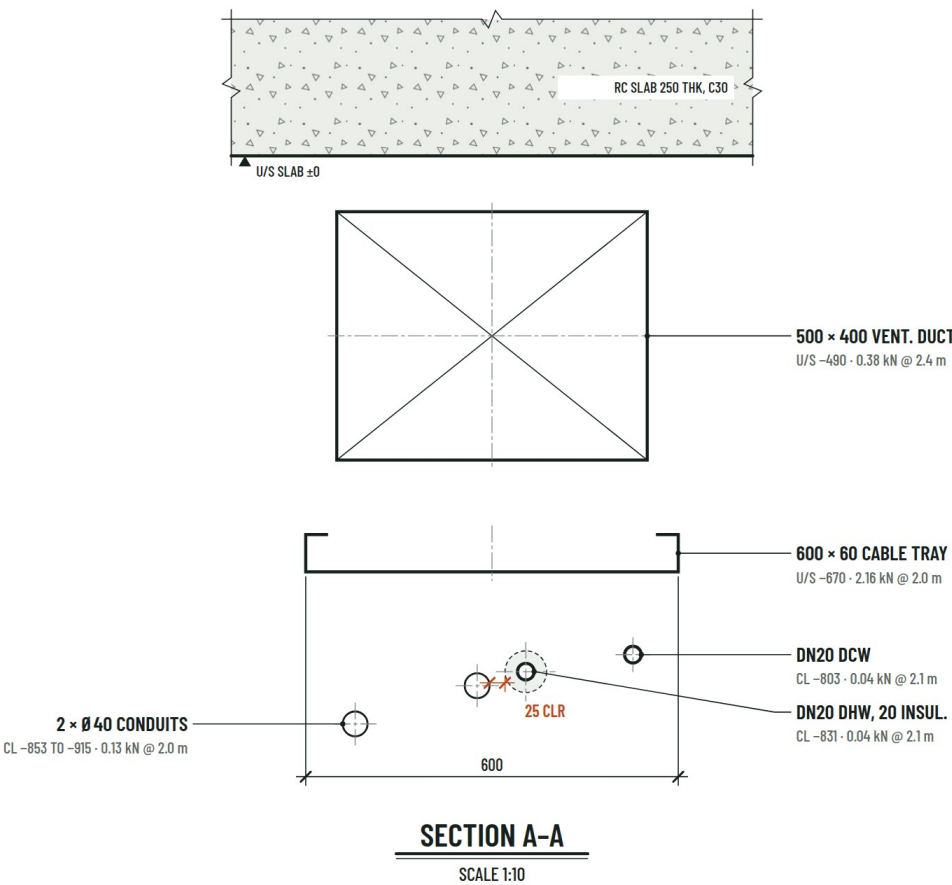


[github.com/lavinia-ped/CrossMEP\\_Dataset](https://github.com/lavinia-ped/CrossMEP_Dataset)  
lavinia@stanford.edu

# Appendix: two more sections from the Generator Studio



Tier C6, seed 0, section 5: six pipes fixed to a wall; offsets from the wall face, the closest clear gap 25 mm.



Exact mix: 2 pipes, 1 tray, 1 duct, 2 conduits on a slab; levels below the soffit and the load at the support.

# Appendix: per-tier statistics, benchmark split

| tier | contexts | median clear gap (mm) | median load (kN) | median width (mm) |
|------|----------|-----------------------|------------------|-------------------|
| C1   | 125      | —                     | 0.10             | 102               |
| C2   | 125      | 120                   | 0.23             | 344               |
| C3   | 125      | 98                    | 0.44             | 613               |
| C4   | 125      | 90                    | 0.55             | 728               |
| C5   | 125      | 73                    | 0.78             | 1027              |
| C6   | 125      | 76                    | 1.21             | 993               |
| C7   | 125      | 60                    | 1.49             | 1138              |
| C8   | 125      | 62                    | 1.71             | 1448              |

4,500 elements

2,529 pipes · 1,141 conduits · 569 trays · 261 ducts

28 mm

Wasserstein-1, generated to measured pipe gaps of the clinic  
(95 % CI 16–44)

–0.36 / +0.56

rank correlation of tier with the closest gap / with the load

# Appendix: the realism check, in detail

| pipe gaps             | pairs | sections | median (mm) | < 25 mm |
|-----------------------|-------|----------|-------------|---------|
| generated (benchmark) | 1,543 | —        | 207         | 0 %     |
| Clinic, Plumbing      | 797   | 3,833    | 203         | 2 %     |
| Duplex, MEP           | 74    | 204      | 177         | 23 %    |
| Duplex, Plumbing      | 41    | 112      | 114         | 32 %    |

- Sections every 250 mm; runs within 400 mm of height form a row; gaps between bare pipe surfaces, below 600 mm
- Robust: across 11 settings of the cut (spacing, row band, axis tolerance, run length) generated ↔ clinic stays at 27–36 mm
- Noise floor: the distance a perfect generator would still show at that sample size
- The duplex samples are small, hence the wide intervals

| Wasserstein-1 (mm)           | by length   | each pair once | noise floor |
|------------------------------|-------------|----------------|-------------|
| generated ↔ Clinic           | 28 (16–44)  | 15 (11–24)     | 8           |
| generated ↔ Duplex MEP       | 70 (57–99)  | 94 (74–119)    | 25          |
| generated ↔ Duplex Plumbing  | 85 (59–132) | 100 (85–125)   | 34          |
| Clinic ↔ Duplex MEP          | 71 (54–101) | 88 (70–118)    | —           |
| Clinic ↔ Duplex Plumbing     | 85 (55–137) | 94 (77–121)    | —           |
| Duplex MEP ↔ Duplex Plumbing | 26 (24–124) | 25 (21–108)    | —           |

Intervals: 95 % bootstrap over contexts (generated) and pipe pairs (measured). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0; derived measurements and attribution in [verify/measured/](#).

# Appendix: the catalog stress test, with error bars

| tier | pipes | benchmark % (95 % CI) | population % (95 % CI) |
|------|-------|-----------------------|------------------------|
| C1   | 71    | 11.3 (4.1–19.1)       | 11.4 (9.6–13.3)        |
| C2   | 147   | 12.2 (6.5–18.8)       | 10.9 (9.4–12.5)        |
| C3   | 203   | 10.3 (5.7–15.6)       | 10.8 (9.6–12.0)        |
| C4   | 257   | 13.6 (8.4–19.3)       | 10.4 (9.4–11.4)        |
| C5   | 391   | 9.2 (6.1–12.6)        | 10.5 (9.6–11.5)        |
| C6   | 410   | 11.5 (7.9–15.4)       | 10.8 (9.9–11.8)        |
| C7   | 465   | 10.8 (7.5–14.1)       | 11.0 (10.1–11.8)       |
| C8   | 585   | 13.3 (9.8–17.1)       | 11.2 (10.4–12.0)       |

| attach diameter rule | bins $\pm 0$ mm | bins $\pm 0.5$ mm | bins $\pm 1$ mm |
|----------------------|-----------------|-------------------|-----------------|
| service (paper)      | 11.6            | 12.9              | 12.9            |
| bare                 | 11.6            | 16.4              | 16.4            |
| insulated            | 6.2             | 7.5               | 7.5             |

% of benchmark pipes attachable when both bin edges are widened by the stated amount. DN100 (114.3 mm) lies 0.3 mm above the 108–114 mm bin, so the headline moves with the diameter rule and the edge tolerance.

| split     | pipes  | % attachable (95 % CI) |
|-----------|--------|------------------------|
| train     | 12,930 | 10.8 (10.2–11.5)       |
| val       | 1,254  | 9.4 (7.4–11.4)         |
| test      | 1,271  | 9.0 (7.2–11.0)         |
| benchmark | 2,529  | 11.6 (10.1–13.0)       |

- Intervals: cluster bootstrap over contexts (2,000 resamples), because the elements of one context are correlated
- Population: 2,000 freshly generated contexts per tier; overall 10.8 % (95 % CI 10.5–11.2)
- Only size and capacity are tested, so coverage is an upper bound on what can be installed
- Best possible share with k sizes: exact optimum over the distinct attach diameters

# Appendix: one record, as stored

```
{
  "context_id": "mep_57", "tier": "C2",
  "surface": {
    "kind": "ceiling", "substrate": "concrete_C25", "thickness_mm": 200.0
  },
  "n_elements": 2, "n_levels": 1, "total_load_kN": 1.12, "bundle_width_mm": 588.0,
  "elements": [
    {
      "kind": "cable_tray", "service": "tray", "trade": "electrical", "shape": "rect",
      "label": "300 tray", "width_mm": 300.0, "height_mm": 60.0, "insulation_mm": 0.0,
      "load_kN": 1.08, "span_m": 2.0, "load_kN_per_m": 0.5396,
      "level": 0, "along_mm": -144.0, "out_mm": 120.0
    },
    {
      "kind": "pipe", "service": "heating", "trade": "heating", "shape": "round",
      "label": "DN20", "width_mm": 26.9, "height_mm": 26.9, "insulation_mm": 20.0,
      "load_kN": 0.04, "span_m": 2.1, "load_kN_per_m": 0.0189,
      "level": 0, "along_mm": 260.5, "out_mm": 156.2
    }
  ]
}
```

- Units: millimetres, kilonewtons, metres (span)
- Widths are bare; insulation\_mm is per side
- $\text{load\_kN} = \text{load\_kN\_per\_m} \times \text{span\_m}$
- along\_mm, out\_mm: position along and out from the surface
- level: the generative row (metadata)